Rockchip

RV1126

A high-performance vision processor SoC for IPC/CVR

Features

CPU

- Quad core ARM Cortex-A7
- MCU

NPU

- 2.0Tops, support INT8/ INT16

Memory

- 32bit DDR3/DDR3L/LPDDR3/DDR4/LPDDR4
- Support eMMC 4.51, SPI Flash，Nand Flash
- Support fast booting

Display

- MIPI-DSI/RGB interface
- 1080P @ 60 FPS

2D Graphics Engine

- Support rotation, x-mirror, y-mirror
- Support alpha blending
- Support scale down/up

Multi-Media

- 14M ISP 2.0 with 3F HDR (Line-based/Frame-based/DCG)
- Support 2*MIPI CSI /LVDS/sub LVDS
- DVP interface with BT.656/BT.1120
- 4K H.264/H.265 30fps video encoder
  - 3840 x 2160@30 fps+720p@30 fps encoding
- 4K H.264/H.265 30fps video decoder
  - 3840 x 2160@30 encoding + 3840 x 2160@30 fps decoding

External Interface

- RGMII interface with TSO network acceleration
- USB 2.0 OTG and USB 2.0 host
- Dual SDIO 3.0 interface for Wi-Fi and SD card
- 8ch I2S with TDM/PDM, 2ch I2S

SDK

- Linux-4.19 base SDK

Physical Specifications

- Power Consumption
  - Typical power consumption at 4K30FPS: 1.8W

Device Information

Package:

- FCCSP409LD
  Body: 14mm x 14mm
  Ball pitch: 0.65mm

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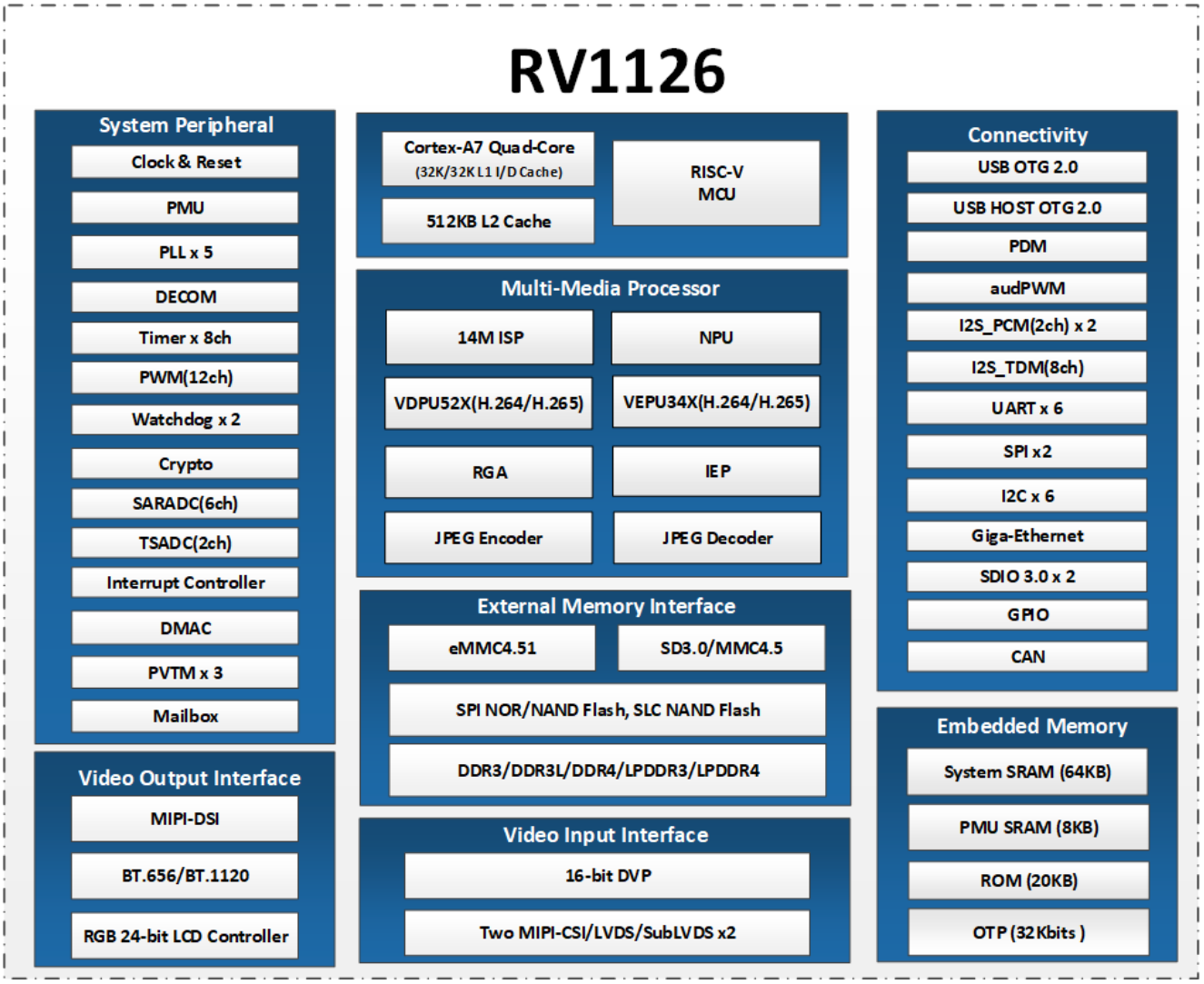
RV1126 Target Applications

- Drone
- Dashboard Camera
- ……

Block Diagram



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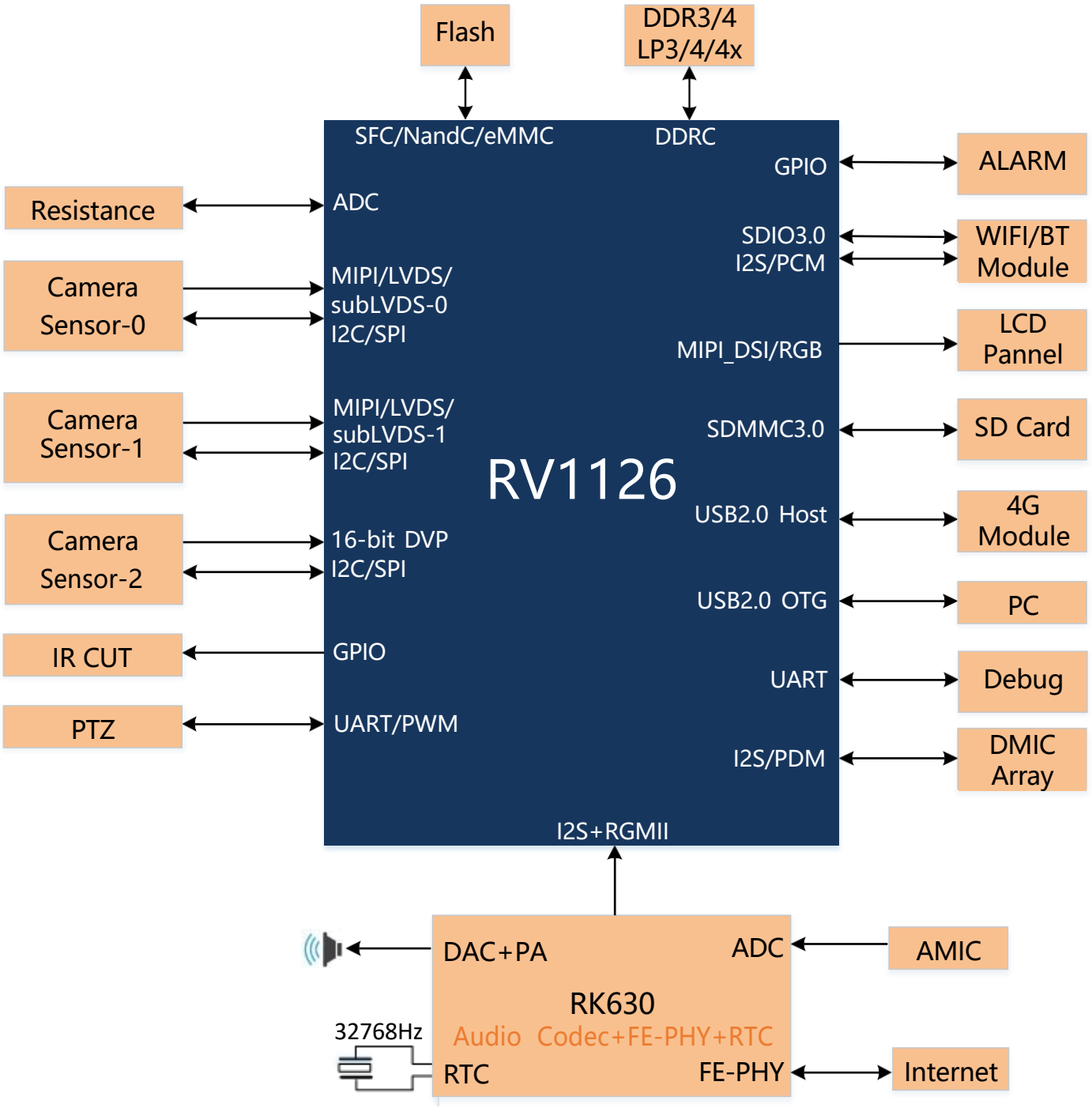
Typical Application Diagram – IPC



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